

SECTION 11522

BULK GAS SYSTEMS

PART 1 - GENERAL

1.1 WORK INCLUDED

- A. This Section specifies the requirements to design, fabricate, deliver, install, start up, validate, and guarantee performance for the stand-alone site and inhouse gas systems, referred in this Section as System.
- B. A single contractor, referred to in this document as Contractor, shall have sole responsibility for providing, installing, and starting up System. Contractor shall have sole responsibility for compliance with the project contract documents; and shall coordinate with other participants in the larger project under construction on the site to complete System according to the project schedule.
- C. The System shall include the equipment of bulk gas piping and specialty gas supply equipment and CDA CQC NOT included bulk gas purifiers.
- D. The System also shall include site bulk gas piping systems from supply interface points at L10 Gas room and/or expansion exist system at the end valves of purifier rooms, Gas qualities control system located in CQC room area L10 of the cub building or other end valve in different level of the fab and distribution to all the areas of usage.
- E. The System shall include distribution of dry air (CDA) from an interface point located in the cub building.
- F. Gas monitoring system.
- G. Gas detector system.

1.2 RELATED WORK

- A. This Section shall be used in conjunction with all other specifications attached in this turn-key package and related Contract Documents to establish the total requirements for System.
- B. CAUTION: Use of this Section without including all of the above-listed items will result in omission of basic requirements.
- C. In the event of conflict regarding the system requirements between this Section and any other section, the provisions of this Section shall govern.

1.3 SYSTEM DESCRIPTION

- A. The System shall be designed, fabricated, installed, and validated in compliance with the requirements of this section and appendix. The System shall be suitable for continuous operation, requiring minimal maintenance and operator intervention.
- B. The System scope includes the following gas subsystems:
 - 1. Inert and non-hazardous specialty gases, including:
 - a. GN2.
 - b. Cylinder racks, cabinet requirement:
 - (1) H2, new Gas cabinet is required.
 - (2) He, N2 new cylinder rack is required.
 - (3) Others inert gases will be expansions from exist VDB/VMB, Rack and/or cabinets as required.
 - c. Distribution tubing and valve manifold boxes.
 - d. Storage cabinet of incoming and outgoing cylinders.
 - e. Cylinder heating system and tubing heat tracing system.
 - f. Purge gas system.
 - g. Fire sprinkler systems internal to the storage, dispensing, and distribution equipment.
 - 2. Flammable gases, including:
 - a. H2 Flammable gases.
 - b. Gas cabinets, Gas cylinder as required.
 - c. Distribution tubing, distribution manifold box and valve manifold boxes.
 - d. Fire sprinkler systems internal to the storage, dispensing, and distribution equipment.
 - e. Storage cabinets of incoming and outgoing cylinders with exhausted containment.
 - f. Purge gas and high-pressure testing system.
 - g. Flammable detector required for dispensing system.
 - 3. Stand-alone control system and gas management system with FMCS interface for monitoring.
 - 4. The utility piping system shall be provided by contractor for supplying qualified utilities to gas equipment as required, include instrument gas, venturi gas, exhaust duct, purge gas, pressure testing piping and sprinkler fire water.
- C. The System scope includes the following bulk gas subsystems:
 - 1. In-house bulk gas piping systems from CUB provided POCs (Points of Connection) to extend the distribution piping including:
 - a. Nitrogen (N2).
 - b. Hydrogen (H2).

- c. Compressed Dry air (CDA).
- D. The system scope includes Gas monitoring system:
 - 1. A control system shall be provided for collecting all the data from the local instruments of gas system, gas detector and the provide remote control to the instrument in hardware control required. Include control panels, terminal panel and wiring as required.
 - 2. The gas monitoring system shall integrate all the gas detectors which installed on gas system and tools.
 - 3. Provide communication net system to integrate all the controllers which installed in the gas system and report to FMCS system.
 - 4. The graphic monitoring system shall be provided and programming for integrating all the information from the controllers of gas cabinet's, gas VMB, gas purifier, CQC system, gas monitoring system and gas yard. Operator is capable to control and monitor whole the gas system through the graphic monitoring system at gas control room.
- E. Provide power supply distribution panels and cabling
 - 1. Gas equipment.
 - 2. Gas monitoring system
 - 3. Income breaker of power panel
- F. In general, single points of connection will be provided for each use point of utility System located. The Contractor shall provide the design and installation from utility points-of-connection as required by the System.
- G. Sub-structural shall be provided as necessary for system installation.
- H. The Contractor shall provide electrical and control systems as necessary to support equipment and piping of the subsystems.
- I. Design, supply, and install piping, pipe supports, seismic restraints, instrumentation and controls, electrical equipment, conduit, wiring, ductwork, and other equipment and materials necessary to provide a complete and working System within the designated areas. Provide catwalks, or raised platforms as necessary to fit within the designated areas.
- J. Provide the common support on the corridors of fab area if required.

1.4 DESIGN CRITERIA

- A. This Section represents the minimum system requirements, and does not include all of the design details required for a fully functional system meeting the operational and performance requirements. It is the responsibility of the Contractor to provide a complete and functional system as

required to meet the intent of the Specifications and to achieve proper performance. The information in this Section and attached appendix shall be used as guidelines of Owner expectations. In the Bid, identify exceptions to this information and proposed alternates.

- B. The System shall convey, purify as required, monitor, dispense, and distribute gases as defined in this Section on a continuous basis, as measured at any sample location on the supply mains. The System shall be capable of producing gas of the quality listed in the attached appendix 100 percent of the time, 24 hours per day, 365 days per year.
- C. Contractor personnel and sub-contractors on-site to aid in installation or maintenance of System shall comply with all rules, regulations, and requirements applicable to those working on the site.
- D. Contractor will be provided space for subsystems and piping installation as designated on the drawings. Additional space requirements shall be identified by contractor in the Bid.
- E. Provide a complete design, including P&IDs, process flow diagrams, pipe and equipment plans and details, equipment skid general arrangements, storage vessel fabrication drawings, equipment support and restraint details, and pipe support and restraint details for systems based on information in Appendix and approved Bid. This shall include intermediate design reviews with Owner. Electronic copies of drawings in Appendix will be provided to Contractor after Contract award. Update these drawings and provide additional drawings as needed. Drawings shall be developed following the project CAD standards for layering, symbology, and other criteria.
- F. Reviews will be carried out at a location to be designated near the Owner's facilities. Three major design reviews, as a minimum, shall be carried out at partial completion milestones of the design. Acceptance of the partially completed design is required as a basis for subsequent design work:
 - 1. Process Development including Process Flow Diagrams along with Material and Energy Consumption and Production.
 - 2. Process Design including Piping and Instrumentation Diagrams along with equipment sizing, piping sizing, component selection, and General Equipment and Piping Arrangements.
 - 3. Detailed Design including Detailed Piping and Support Drawings; equipment, materials, and component selection; and utility and accessory system designs.
- G. General grade bulk gas system piping shall be bright annealed 316L (mechanical polished Ra25 average) purified

bulk gas system piping shall be Electrical polish (EP) 316L, the DA system shall be SS304AP welded grade tube.

- H. Contractor shall be responsible for coordinating System design and construction with the facilities and utilities provided by the Owner, and with other site construction activities through weekly meetings with the designated Space Management (SM) Coordinator.
- I. The System and its operations shall be designed to maximize energy and gas conservation and to minimize the use of chemicals and generation of hazardous waste.
- J. The system must be designed of capable maintenance and system expansion under the condition of non-shutoff of power and system. (系統須符合在不斷電，不停機狀況下，進行擴充或歲修。)

1.5 WARRANTY

- A. System warranty shall comply with the project requirements identified elsewhere in the bid documents. In addition, warranties provide to Contractor by OEMs that exceed the minimum project requirements, shall be identified in the bid and passed-on to the Owner.

1.6 REGULATORY REQUIREMENTS

- A. Obtain permits and arrange for necessary inspections required by codes applicable to System. Submit written evidence that the required permits, inspections, and code requirements have been secured prior to any on-site work being performed.

1.7 QUALITY ASSURANCE/QUALITY CONTROL

- A. A quality assurance/quality control program (QA/QCP) shall be established according to the requirement in the section 01001B-A QUALITY CONTROL representative, and maintained throughout the project. The QA/QCP shall be clearly documented in the Bid and is subject to review, acceptance, and monitoring by the Engineer and Owner during the course of project execution.
- B. The QA/QCP shall include, at a minimum:
 - 1. Material handling.
 - 2. Incoming material and equipment inspections.
 - 3. Rejection and/or replacement of material.
 - 4. Pipe Fitter training and performance in the shop and in the field.
 - 5. Fabrication and installation procedures.
 - 6. Design compliance checks.
 - 7. Monitoring of submittal submissions and comment

- incorporation.
- 8. Development of the detailed startup and validation plan for the System.
- 9. Coordination between the Contractor and site installation activities.
- 10. Coordination between the Contractor and the Engineer.
- 11. Coordination of multiple disciplines within the Contractor's organization (i.e. electrical, I&C, shop fabrication, mechanical).
- C. A QA/QC manager (QA/QCM) shall be designated to administer the Contractor's QA/QCP. The QA/QCM shall have at least 5 years' experience in the implementation of high-purity piping and bulk gas systems.
- D. The Owner and/or Engineer may review the Contractor's facilities or fabrications in process at any time.
- E. The Owner and/or Engineer reserves the right to review equipment prior to shipping and to witness factory testing.
- F. Notify the Owner at least 3 weeks before scheduled ship dates of equipment.
- G. Owner shall be invited to attend all onsite piping tests, hardware tests, component tests, system tests, and functional demonstrations and be informed of such tests or demonstrations at least 3 days in advance.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Deliver materials to the jobsite and provide storage space on or off site as necessary. Coordinate delivery schedules, storage, construction, and laydown areas with the Owner. Provide cranes, Labor, and material-handling equipment required to unload, transport, and protect equipment on the jobsite.

1.9 SUBMITTALS

- A. Provide submittals as identified by the submittal and design deliverables checklist in the attached appendix.

1.10 PROJECT RECORD DOCUMENTS

- A. After system installation, provide complete, updated drawings showing the field-verified, as-built condition of the system, as required by the attached submittal and design deliverables checklist in the attached appendix. All submitted documents and drawings shall be submitted in both hard copy and soft copy (burn onto a CD) format.

1.11 OPERATION AND MAINTENANCE DATA

- A. The O&M manuals shall include the information identified by the attached submittal and design deliverables checklist in the attached appendix.

1.12 OPERATIONS AND MAINTENANCE SERVICE

- A. Operate and maintain the System until it is completed, validated, and accepted by the Owner. If the System or portions of the System are not completed and accepted by the Owner by the contracted scheduled dates provide personnel at the Contractor's expense to operate the System and provide gases as needed to support the fab startup activities until the System is completed and accepted by the Owner.
- B. Provide operations and maintenance services for System as Bid options, and as identified by the attached Submittal and Design Deliverables Checklist in Appendix.

PART 2 - PRODUCTS

2.1 GENERAL

- A. Equipment and component Labeling shall be in compliance with the contract.
- B. System electrical and controls systems shall be in compliance with requirements of the contract documents.
- C. Facility piping supports shall comply with the requirements of the contract documents.
- D. System noise and vibration considerations shall be provided according to requirements of the contract documents.
- E. A minimum of three lifting lugs shall be provided for each piece of equipment weighing more than 100 kilograms.
- F. Pressure vessels shall be fabricated according to the requirements of the ASME Boiler and Pressure Code, Section VIII, Division 1; or Japanese or European code equivalent; and shall carry an ASME, or Japanese or European code equivalent, code stamp.
- G. Equipment, piping, conduit, and other installation shall be installed, anchored, and braced to withstand site seismic design requirements. Anchoring and bracing design for compliance with seismic requirements shall be done by a licensed professional engineer, and calculations substantiating the design and including the stamp of the licensed professional engineer shall be submitted. Such installation shall comply with the requirements of the applicable codes and standards, and the terms of the contract.
- H. Equipment placement shall allow safe and convenient

- access to equipment, valves, and instrumentation requiring maintenance and calibration.
- I. Equipment selection, cabinets, sound-insulating materials, and other sound deadening techniques shall be used to limit noise levels to 85 dBA, as measured on the "A" scale of a standard sound level meter at slow response at 1 meter from each piece of equipment, pipe, or valve during operation.
 - J. Programming for the PLCs and other programmable devices shall be provided by Contractor. The programming shall be done in a manner consistent with the project programming standards. The project programming standards will be provided by the project FMCS vendor.
 - K. Coordinate with the project FMCS vendor. Contractor is responsible for verifying that I/O assignments and databases are uniquely identified within the site facility monitoring and control system and that they comply with the facility tagging standards. Verify that System and control points are capable of being monitored from the facility monitoring and control system, as required.

PART 3 - EXECUTION

3.1 INSPECTION

- A. Provide the shop fabrication testing prior to equipment shipment as required by contract documents.

3.2 INSTALLATION

- A. Provide onsite construction offices, fabrication shops, temporary high-purity pipe cleanrooms, and worker facilities to support System installation in coordination with the Owner and with other site construction activities.
Key milestone completion dates shall be as set by the Owner. Installation schedules and construction activities shall be developed by Contractor and coordinated with Owner. Attend daily and weekly coordination meetings as required by Owner.
- B. Prepare and submit complete installation plans, showing equipment location, piping, and wiring details for review and approval by Owner. Installation shall be according to the approved plans. Deviations will not be permitted unless approved in advance by the Owner.

3.3 COORDINATION

- A. Verify with Owner the other construction activities scheduled concurrent with system installation activities. Coordinate work schedule for a minimum of interference with work of other trades. Coordinate with installers for other systems to define space requirements and clearance requirements with respect to other equipment and components. Owner reserves the right to determine space priorities where interference occurs.
- B. Verify locations of under slab utilities and other existing site constraints prior to finalizing gas system designs. The Contractor shall field-verify the existing site layouts, obstructions, and equipment/piping configurations.

3.4 ADJUSTING AND CLEANING

- A. Furnish expendables and consumables needed for startup, qualification, and operation of the System for the qualification period prior to System acceptance.

3.5 DEMONSTRATION

- A. Piping shall be tested according to the requirements of ANSI B31.3 and the contract documents.
- B. Provide a minimum of 1 week of training for the Owner's personnel. Training shall consist of a combination of classroom and in-the-field training on the System operational and monitoring capabilities for an overall understanding of the System design, piping distribution, monitoring and control capabilities, and utility requirements. Agenda shall be submitted to Owner for approval. Where necessary for proper instruction, equipment manufacturer's personnel shall be brought in for instruction on particular equipment. The Owner will establish the actual training dates at a later date.
- C. Formal acceptance of the System by the Owner will occur using a certificate of substantial completion. The certificate will not be issued and the System not accepted until the following have been proven by the Contractor:
 - 1. System operational performance, and System product quality. The System shall consistently produce gas in compliance with the requirements of this Section and appendix for four continuous weeks prior to System acceptance.
 - 2. Completed Quality Assurance Program (QAP) report.
 - 3. Documentation of operational and functional acceptance tests.
 - 4. Completion of punch list items.
 - 5. Submittal of items identified by the Submittal and

Design Deliverables Checklist in the attached appendix.

6. Completion of training classes.

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